Twin–Cousin Goldbach Representations

Four-Prime Block Certificates, Modular Restrictions, and Cumulative Rescue Dynamics

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21 October 2026

Research note — conjectural statements distinguished from proven results

Abstract

We formulate two related conjectures about representations of even integers by genuine prime numbers belonging to twin- and cousin-prime pairs. The *Twin–Cousin Goldbach Conjecture* (TCG) asks for a representation of each even integer at least six as a sum of a twin prime and a cousin prime. The stronger *Twin–Cousin Block Conjecture* (TCB) requires four primes in two prescribed pairs that simultaneously represent four consecutive even integers. We prove the implication TCB $\Rightarrow$ TCG, establish exact prime-arrival times in a cumulative Goldbach filtration, derive a finite rescue deadline and an equivalent frontier-buffer criterion, and calculate the complete local divisibility obstruction and its singular series. A congruence restriction modulo five explains a marked asymmetry in the observed rescue process. Explicit deterministic verification establishes TCB for block indices $2 \le m \le 5000$, while the dynamic calculation is extended to stage $n = 15000$. No unconditional proof of either conjecture is claimed.

Keywords: Goldbach-type problems; twin primes; cousin primes; prime quadruples; admissible linear forms; singular series; arithmetic sieves.
MSC 2020: 11P32, 11N36, 11A41.

1 Introduction

Goldbach's binary problem and the Hardy–Littlewood predictions for prime patterns are central reference points for questions about simultaneous primality [3]. Here we constrain both prime summands to belong to specific, short prime pairs. The resulting question is not merely whether a given even integer is representable, but whether the accumulated supply of *actual* twin and cousin primes is sufficient to certify successive targets.

The purpose of this note is to isolate two conjectures, develop exact identities and a deterministic cumulative construction around them, and state clearly which assertions have been established and which remain open. The block formulation is particularly useful: a single four-prime certificate yields four Goldbach representations and can repair several adjacent omissions simultaneously. The admissibility and singular-series calculations belong to the general tradition of prime-tuple heuristics and sieve theory [4, 2, 1]; these heuristics are not invoked as proofs of prime existence.

2 Two conjectures and their logical relationship

Definition 2.1 (Twin and cousin prime membership)**.** An odd prime t is a *twin prime* if at least one of $t-2, t+2$ is prime. An odd prime c is a *cousin prime* if at least one of $c-4, c+4$ is prime. We allow $t = c$ and do not demand that the prime pairs be disjoint.

Conjecture 2.2 (Twin–Cousin Goldbach Conjecture, TCG)**.** *Every even integer $N \geq 6$ possesses a representation*

$$N = t + c, \tag{2.1}$$

where t is a twin prime and c is a cousin prime.

Except for $(3, 5)$, each twin-prime pair has the form $(6a - 1, 6a + 1)$, $a \geq 1$. Except for $(3, 7)$, each cousin-prime pair has the form $(6b + 1, 6b + 5)$, $b \geq 1$. These are the *regular pairs* below. The exceptional pairs remain available for TCG.

Definition 2.3 (A four-prime certificate)**.** For integers $a, b \geq 1$, call (a, b) a *block certificate* if all four numbers

$$6a - 1, \qquad 6a + 1, \qquad 6b + 1, \qquad 6b + 5 \tag{2.2}$$

are prime. Its *block index* is $m = a + b$.

Conjecture 2.4 (Twin–Cousin Block Conjecture, TCB)**.** *For every integer $m \geq 2$, there exist $a, b \geq 1$, with $a + b = m$, for which (2.2) are all prime.*

Proposition 2.5 (Four-number certificate identity)**.** *A block certificate (a, b), $a + b = m$, supplies twin–cousin representations for all four even integers*

$$B_m := \{6m, 6m + 2, 6m + 4, 6m + 6\}. \tag{2.3}$$

Proof. Put $p = 6a - 1$ and $q = 6b + 1$. Since $p, p + 2$ are twin primes and $q, q + 4$ are cousin primes, their four cross-pair sums are

$$p + q = 6m, \quad (p + 2) + q = 6m + 2, \quad p + (q + 4) = 6m + 4, \quad (p + 2) + (q + 4) = 6m + 6.$$

Each uses one prime of each permitted kind. □

Theorem 2.6 (Logical implication)**.** *The* TCB *conjecture implies* TCG.

Proof. If every B_m for $m \geq 2$ is certified, their union covers every even integer from 12 onward: adjacent blocks overlap at $6m + 6 = 6(m + 1)$. The three remaining even integers have the permitted representations

$$6 = 3 + 3, \qquad 8 = 5 + 3, \qquad 10 = 3 + 7.$$

Here $3, 5$ belong to the twin pair $(3, 5)$, and $3, 7$ to the cousin pair $(3, 7)$. □

Remark 2.7. *The converse is* not established. *TCG permits a separate representation for each even integer and also admits the exceptional pairs. TCB insists that a single pair of regular prime pairs certify an entire quartet. In particular, failure to certify B_m does not automatically mean that every member of B_m lacks an individual twin–cousin representation.*

3 Exact certificate counts and modular obstructions

For $m \geq 2$ define the *certificate count*

$$R(m) := \sum_{a=1}^{m-1} \mathbf{1}_{\mathbb{P}}(6a - 1)\mathbf{1}_{\mathbb{P}}(6a + 1)\mathbf{1}_{\mathbb{P}}(6(m - a) + 1)\mathbf{1}_{\mathbb{P}}(6(m - a) + 5), \tag{3.1}$$

where $\mathbf{1}_{\mathbb{P}}$ denotes the prime indicator. Thus TCB is exactly the assertion $R(m) > 0$ for every $m \geq 2$. Formula (3.1) counts certificates, not the total number of individual Goldbach representations.

Lemma 3.1 (Forbidden residues)**.** *Fix $m \geq 2$ and a prime $\ell \geq 5$. Write $u \equiv 6^{-1} \pmod{\ell}$. Among the four linear expressions*

$$L_1(a) = 6a - 1,\ L_2(a) = 6a + 1,\ L_3(a) = 6(m-a) + 1,\ L_4(a) = 6(m-a) + 5, \tag{3.2}$$

the forbidden residue classes of $a \pmod{\ell}$ are

$$u, \quad -u, \quad m + u, \quad m + 5u. \tag{3.3}$$

If $\rho_\ell(m)$ denotes the number of distinct forbidden classes, then

$$\rho_\ell(m) = \begin{cases} 3, & \ell \mid m(m+1)(3m+1)(3m+2), \\ 4, & \text{otherwise.} \end{cases} \tag{3.4}$$

For $\ell = 2, 3$, none of the four expressions ever vanishes modulo ℓ.

Proof. Solving $L_j(a) \equiv 0 \pmod{\ell}$ yields (3.3). Any equality of distinct listed residues implies one of the conditions $m \equiv 0, -1 \pmod{\ell}$ or $3m + 1 \equiv 0,\ 3m + 2 \equiv 0 \pmod{\ell}$; conversely each produces an equality. For $\ell \geq 5$ these four conditions are mutually exclusive, so at most one collision occurs. Finally, each L_j is congruent to ± 1 modulo 2 and to ± 1 modulo 3. □

Theorem 3.2 (Uniform local admissibility)**.** *For every $m \geq 2$ and every prime ℓ, there is a class $a \pmod{\ell}$ on which none of the four forms (3.2) is divisible by ℓ. In particular, the four-form family has no fixed local prime obstruction.*

Proof. For $\ell = 2, 3$ every class is permissible. For $\ell \geq 5$, Lemma 3.1 gives $\rho_\ell(m) \leq 4 < \ell$. Hence at least one residue survives. □

Corollary 3.3 (Positive singular series)**.** *The associated Hardy–Littlewood singular series is*

$$\mathfrak{S}(m) = 81 \prod_{\substack{\ell \geq 5 \\ \ell\ mathrmprime}} \frac{1 - \rho_\ell(m)/\ell}{(1 - 1/\ell)^4}. \tag{3.5}$$

It converges for every fixed $m \geq 2$ and satisfies the uniform lower bound

$$\mathfrak{S}(m) \geq S_0 := 81 \prod_{\substack{\ell \geq 5 \\ \ell\ mathrmprime}} \frac{1 - 4/\ell}{(1 - 1/\ell)^4} > 0. \tag{3.6}$$

Proof. At 2 and 3 there are no roots; the two local factors multiply to $16(3/2)^4 = 81$. For $\ell \geq 5$, replacing $\rho_\ell(m) \in \{3, 4\}$ by 4 lowers the local factor. The resulting generic factor is $1 + O(\ell^{-2})$ and is strictly positive, so its infinite product converges to a positive number. For fixed m, only finitely many primes divide the fixed nonzero product in (3.4), ensuring convergence of (3.5). □

Proposition 3.4 (The restrictive residue class modulo five)**.** *If $m \equiv 2 \pmod{5}$ and $(a, m-a)$ is a block certificate with $a \geq 2$, then $a \equiv 0 \pmod{5}$. For each other class of $m \pmod{5}$, two residue classes of $a \pmod{5}$ are locally admissible; for $m \equiv 2$ only one survives.*

Proof. Modulo five the forbidden residues are $1, -1, m+1, m$. Their complement is $\{0\}$ when $m \equiv 2 \pmod{5}$, and has size two otherwise. If $a \geq 2$, the four numbers in (3.2) all exceed five except possibly $6a - 1$ when $a = 1$; thus a certificate cannot exploit a factor equal to the prime 5. The restriction follows. □

Remark 3.5. *The exception $a = 1$ is genuine, not an artefact: $(a, b) = (1, 21)$ certifies $m = 22$ using the primes $5, 7, 127, 131$. The modulo-five restriction concerns potential* composite divisibility*, not an absolute ban on a value equal to five. The local factor at five for $m \equiv 2$ is half that for the other four residue classes.*

4 Cumulative prime arrivals and rescue dynamics

Fix an integer stage $n \geq 3$. Call a prime *observed by stage n* if it occurs as an addend in a Goldbach decomposition of at least one of $6, 8, \ldots, 2n$.

Lemma 4.1 (Exact observation cutoff)**.** *The observed primes at stage n are precisely the odd primes $p \leq 2n-3$.*

Proof. If $p \leq 2n-3$ is an odd prime, then $p+3 \leq 2n$ is even, and $p+3 = p+3$ is a Goldbach representation. Conversely, every Goldbach addend of an even number at most $2n$ is at most $2n-3$, as the other odd prime is at least three. □

Proposition 4.2 (Arrival times)**.** *A regular twin pair $(6a-1, 6a+1)$ is fully observed first at stage $3a+2$. A regular cousin pair $(6b+1, 6b+5)$ is fully observed first at stage $3b+4$. Hence the certificate (a, b) arrives at*

$$\tau(a,b) = \max\{3a+2,\ 3b+4\}. \tag{4.1}$$

If $a+b=m$, then $\tau(a,b) \leq 3m+1$.

Proof. The cutoff of Lemma 4.1 must reach the larger member of each pair. Solving $6a+1 \leq 2n-3$ and $6b+5 \leq 2n-3$ gives the asserted stages. The deadline follows from $1 \leq a, b \leq m-1$. □

Definition 4.3 (First certification and first hole)**.** Let τ_m be the minimum of $\tau(a,b)$ over all block certificates with $a+b=m$, taking $\tau_m = +\infty$ if no such certificate exists. Let

$$H(n) = \min\{m \geq 2 : \tau_m > n\}, \qquad D(n) = 3H(n) + 1 - n. \tag{4.2}$$

Here $H(n)$ is the first *uncertified block index*, and $D(n)$ is its deadline buffer. These are not defined as the first missing individual Goldbach representation.

Theorem 4.4 (Exact frontier criterion)**.** *The following two statements are equivalent:*

(i) *The Twin–Cousin Block Conjecture holds for every $m \geq 2$.*

(ii) *$D(n) > 0$ at every integer stage $n \geq 7$.*

Whenever these conditions hold, every even target $2n$ is twin–cousin representable.

Proof. Assume (i), and take $m = H(n)$. Since this block has not arrived, $n < \tau_m \leq 3m+1$, by Proposition 4.2; hence $D(n) > 0$. Conversely, assume (ii), and fix $m \geq 2$. At stage $n = 3m+1$, the inequality $3H(n)+1-n > 0$ implies $H(n) > m$, so block m has been certified. Therefore (i) holds. If $n \geq 7$, then $D(n) > 0$ implies $n \leq 3H(n)$. The blocks $B_2, \ldots, B_{H(n)-1}$, all certified, cover the even numbers from 12 through $6H(n)$, which include $2n$ (and $H(n) \geq 3$ for $n \geq 7$). The smaller cases follow from the representations in Theorem 2.6 and $12 = 5+7$. □

Proposition 4.5 (Frontier recurrence and critical rescue)**.** *Write $\Delta H(n) = H(n+1) - H(n) \geq 0$. Then*

$$D(n+1) = D(n) - 1 + 3\Delta H(n). \tag{4.3}$$

If $D(n) = 1$, maintaining $D(n+1) > 0$ requires certification of the current earliest missing block at the next stage.

Proof. Subtract (4.2) at consecutive stages. If $D(n) = 1$ and $\Delta H(n) = 0$, then $D(n+1) = 0$. Conversely a positive integer $\Delta H(n)$ restores positivity. □

Remark 4.6 (Arrival clock)**.** *Apart from the small exceptional primes, stage* $n \equiv 0 \pmod 3$ *introduces no new prime:* $2n-3$ *is a composite multiple of three. At stage* $3k+1$*, a new cousin pair can complete; at stage* $3k+2$*, a new twin pair can complete. These events can share the prime* $6k-1$*. This observation provides a deterministic update schedule but does* not *by itself prove that a rescue must occur at a critical deadline.*

5 Examples of simultaneous rescue

Example 5.1 (Closing 62 and 64 at stage 23)**.** The newly observed twin pair $(41, 43)$, with the already observed cousin pair $(19, 23)$, provides the certificate $(a, b) = (7, 3)$, $m = 10$:

$$60 = 41 + 19, \quad 62 = 43 + 19, \quad 64 = 41 + 23, \quad 66 = 43 + 23.$$

The two missing interior even numbers are repaired at the same stage.

Example 5.2 (Rescue in the restrictive residue class)**.** For $m = 22 \equiv 2 \pmod 5$, the certificate $(a, b) = (5, 17)$ uses the primes

$$29, \quad 31, \quad 103, \quad 107,$$

and arrives at stage $\max(17, 55) = 55$. It gives

$$132 = 29 + 103, \quad 134 = 31 + 103, \quad 136 = 29 + 107, \quad 138 = 31 + 107.$$

A second certificate, $(a, b) = (1, 21)$, uses $(5, 7)$ and $(127, 131)$ and arrives later, at stage 67. Stage 67 is the deadline $3m + 1$.

Example 5.3 (One arrival, several old holes)**.** The twin pair $(71, 73)$ first arrives at stage 38. With cousin pairs $(19, 23)$ and $(43, 47)$, it certifies the distinct block indices $m = 15$ and $m = 19$. An incoming twin pair can therefore perform several repairs at once.

6 Computational evidence

We used an ordinary Eratosthenes sieve to determine primality through 30,010, enumerated all regular twin and cousin pairs in that range, and tested $R(m)$ in (3.1) for every $2 \le m \le 5000$. For such m, no candidate exceeds $6(m-1)+5 \le 29{,}999$, so this bound is sufficient for an exhaustive finite test. The calculation was independently expressed as a stage-by-stage arrival process through $n = 15000$, where the observed prime cutoff is 29,997. An accompanying short Python program reproduces the tabulations.

Proposition 6.1 (Finite verification, computational)**.** *Direct primality testing yields* $R(m) > 0$ *for every integer* $2 \le m \le 5000$*. The associated deadline buffer remains strictly positive for every computed stage* $7 \le n \le 15000$*.*

Verification method. For each m, inspect $a = 1, \dots, m-1$, put $b = m - a$, and increment $R(m)$ precisely when the four sieve entries in (2.2) are prime. To calculate arrival times, record the minimum $\tau(a, b)$ among the successful pairs for each m. Updating $H(n)$ from these minima gives $D(n)$ exactly by (4.2). This is a finite exhaustive procedure; the statement is computer-verified, not an analytic theorem extending beyond the stated bounds. □

The observed dominance of residue class 2 in the frontier waiting times is consistent with Proposition 3.4: fewer locally admissible index choices make certificates rarer. The data do not establish a distribution law, an independence model, or a universal bound on waiting times.

Table 1: Selected stages: the first uncertified block and its remaining rescue buffer. The entry $6H + 2$ is the first interior even target *not certified by the regular block mechanism*; exceptional-pair representations are accounted for separately.

Stage n	$H(n)$	$6H(n)+2$	$D(n)$
22	10	62	9
23	12	74	14
25	15	92	21
32	15	92	14
37	15	92	9
38	20	122	23
43	22	134	24
55	31	188	39
100	54	326	63
500	225	1352	176
1000	477	2864	432
3000	1562	9374	1687
5000	2712	16274	3137
10000	5647	33884	6942
15000	8107	48644	9322

Table 2: Individual twin–cousin sums in selected finite filtrations. The range ends at the largest obtainable even sum using the observed primes; gaps are the missing even integers *within* that range. These are actual missing representations, not merely absent four-prime block certificates.

Stage n	Largest obtainable sum	Gaps in the range $6, 8, \ldots$
22	72	62, 64
23	84	74, 76
25	90	none
32	108	92, 94
37	132	92, 94, 116, 118, 120, 122, 124
38	144	122, 124, 134, 136
40	144	122, 124, 134, 136

7 A heuristic and a precise conditional route

The local computation suggests a Hardy–Littlewood-type expectation

$$R(m) \quad \text{typically of order} \quad \mathfrak{S}(m)\frac{m}{(\log m)^4}. \tag{7.1}$$

There are $m - 1$ candidate values of a and four primality requirements, with the singular series correcting for arithmetic dependence. The strictly positive lower bound (3.6) explains why there is no *local* reason for permanent extinction. It does not prove that any candidate is prime in the *finite* interval $1 \le a < m$. In particular, standard fixed-pattern prime-tuple conjectures must not be silently treated as estimates uniform in the moving parameter m.

For clarity, the following conditional implication states exactly what degree of uniformity would suffice.

Proposition 7.1 (A sufficient analytic hypothesis)**.** *Fix $0 < \delta < 1/2$. Suppose the number $R_\delta(m)$ of certificates with $\delta m \le a \le (1-\delta)m$ satisfies, uniformly as $m \to \infty$,*

$$R_\delta(m) = (1 + o(1))(1 - 2\delta)\mathfrak{S}(m)\frac{m}{(\log m)^4}. \tag{7.2}$$

Table 3: Effect of m mod 5 in the finite computations. Mean certificate counts use indices $50 \le m \le 5000$. The frontier-stage column counts the number of stages $50 \le n \le 10000$ for which $H(n)$ has the indicated residue; successive equal frontier indices contribute repeatedly.

$m \bmod 5$	Mean $R(m)$	Stages with $H(n) \equiv m \pmod 5$
0	30.60	905
1	30.30	251
2	15.88	8424
3	30.93	130
4	32.18	241

Then TCB holds for all sufficiently large m. If its remaining finitely many cases also hold, TCB and hence TCG hold universally.

Proof. By Corollary 3.3, $\mathfrak{S}(m) \ge S_0 > 0$. The right-hand side of (7.2) therefore eventually exceeds one, and in particular is positive. Since $R(m) \ge R_\delta(m)$, there is a certificate for every sufficiently large m. □

Remark 7.2. *Proposition 7.1 is conditional and is not a proof of (7.2). Controlling four simultaneous prime values of this shape is the outstanding analytic difficulty. Admissibility, finite verification, and the exact rescue recurrence each fall short of that step.*

8 Research directions

Three questions arise directly from the proven structure.

(i) **The critical-deadline step.** Can one show that whenever the buffer reaches its last positive unit, some newly available genuine twin or cousin pair certifies the earliest missing block? Proposition 4.5 says exactly what an induction would need; it does not guarantee the required primes.

(ii) **The class $m \equiv 2 \pmod 5$.** Can the forced residue $a \equiv 0 \pmod 5$, together with the exceptional certificate $a = 1$, be exploited to obtain a rigorous certificate count or a sharper finite-interval lower bound?

(iii) **Multiple rescues from a single arrival.** Can the correlations produced by a newly arriving prime pair, which repairs several historic block indices, be used to bound how far $H(n)$ can lag behind stage n? Such a bound must come from actual prime occurrences, not merely candidate congruence classes.

9 Conclusion

TCG is a Goldbach-type problem with two genuine prime-family restrictions. TCB is a stronger, explicitly four-prime assertion. Their relationship, the complete quartet identity, the prime observation cutoff, the rescue deadline, the buffer recurrence, and the complete local root-count formula are proved here. The computations support both conjectures on substantial finite ranges and isolate a particularly restrictive congruence class modulo five. The open step remains a global existence theorem for at least one four-prime certificate at every block index. No unconditional induction across all indices is obtained.

A Reproducibility and range of the finite check

An accompanying file, `verify_twin_cousin.py`, implements the sieve-and-certificate algorithm. In pseudocode, the two essential parts are:

1. Sieve every integer through 30,010. Record regular twin indices a for which $6a \pm 1$ are both prime and cousin indices b for which $6b + 1, 6b + 5$ are both prime.

2. For each pair of recorded indices, set $m = a + b$, increase $R(m)$, and update τ_m by the arrival formula (4.1). For each stage n, scan upward to the first m with $\tau_m > n$; compute $D(n)$. To tabulate actual available even sums, independently include the exceptional pairs $(3, 5)$ and $(3, 7)$ and recombine every eligible twin-prime member with every eligible cousin-prime member.

The prime cutoff is sufficient for verifying all block indices through $m = 5000$. The finite stage computation through $n = 15000$ is independent of any unproved prime-distribution estimate. A computer verification cannot supply a universal prime-existence proof.

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